

Database Architecture Report

Beneficiary & Scheme Disbursement System

This document outlines every SQL feature utilized in your schema, providing a clear definition of what each component is and the architectural reasoning behind why it was chosen.

1. Core SQL Commands

Command	Definition (What it is)	Purpose (Why you used it)
CREATE DATABASE	Creates a new, empty database.	To establish an isolated environment (<code>government_subsidy_db</code>) for this specific project.
USE	Selects a specific database to work within.	Ensures all subsequent table creation scripts are executed inside the newly created database.
CREATE TABLE	Defines a new table, its columns, and rules.	To structure your data logically into distinct entities (e.g., <code>country</code> , <code>beneficiary</code> , <code>scheme</code>).
CREATE INDEX	Builds a searchable data structure on a column.	To significantly speed up search queries (e.g., finding a beneficiary by mobile number).

2. Data Types

Specific data types are used to optimize storage space and ensure the data behaves correctly in a Java/Spring Boot environment.

Data Type	Definition (What it is)	Purpose (Why you used it)
BIGINT	A large whole number (up to 19 digits).	Used for primary IDs to future-proof the system. Subsidies can scale to millions of records.

Data Type	Definition (What it is)	Purpose (Why you used it)
VARCHAR(n)	A variable-length character string.	Used for names, emails, and codes. Saves memory by only using the space needed.
TEXT	A string that can hold a vast amount of text.	Used for <code>description</code> and <code>failure_reason</code> where content length is unpredictable.
DATE	Stores a date (YYYY-MM-DD) without time.	Used for <code>date_of_birth</code> and <code>start_date</code> where the exact hour/minute is irrelevant.
TIMESTAMP	Stores date and exact time.	Used for <code>created_at</code> and <code>updated_at</code> to track exactly when records are created or modified for auditing.
BOOLEAN	A true/false value.	Used for flags like <code>is_active</code> or <code>bank_verified</code> for simple binary logic.
ENUM	A string object chosen from a predefined list.	Enforces strict validation for columns like <code>status</code> or <code>scheme_type</code> .
DECIMAL(p,s)	Exact precision numbers (p=total digits, s=decimals).	Used for <code>amount</code> and <code>score</code> . Crucial for financial data to prevent rounding errors.
JSON	Stores structured JavaScript Object Notation data.	Used in <code>audit_log</code> to flexibly store "before" and "after" states of rows without complex schemas.

3. Database Constraints

Constraints are rules applied to columns to ensure data accuracy and reliability (Data Integrity).

Constraint	Definition (What it is)	Purpose (Why you used it)
PRIMARY KEY	Uniquely identifies each row in a table.	Ensures no two records have the same ID. Essential for Hibernate/JPA mapping.
AUTO_INCREMENT	Automatically generates the next sequential number.	Relieves the application from having to calculate the next ID.
FOREIGN KEY	Links a column to the primary key of another table.	Enforces relational integrity (e.g., cannot add a village to a non-existent <code>block_id</code>).
NOT NULL	Ensures a column cannot be left empty.	Used for mandatory fields to prevent application crashes caused by missing data.
UNIQUE	Ensures all values in a column are different.	Used on <code>beneficiary_uid</code> , <code>mobile_number</code> , etc., to prevent duplicate registrations.
DEFAULT	Assigns a fallback value if none is provided.	Used for <code>is_active = TRUE</code> or <code>status = 'PENDING'</code> to simplify insertion logic.

4. Advanced & Automated Behaviors

ON UPDATE CURRENT_TIMESTAMP

- **What it is:** An automatic trigger built into MySQL.
- **Why you used it:** When applied to `updated_at`, the database automatically updates this timestamp whenever a row is modified, saving backend effort.

Composite Unique Keys (UNIQUE KEY uk_...)

- **What it is:** A uniqueness rule applied to a *combination* of two or more columns.
- **Why you used it:** Ensures a single beneficiary cannot enroll in the exact same scheme twice, preventing duplicate disbursements.

Normalization (Hierarchical Design)

- **What it is:** Breaking down data into smaller, related tables to avoid redundancy.
- **Why you used it:** By separating locations into country > state > district > block > village, consistency is maintained across millions of records.